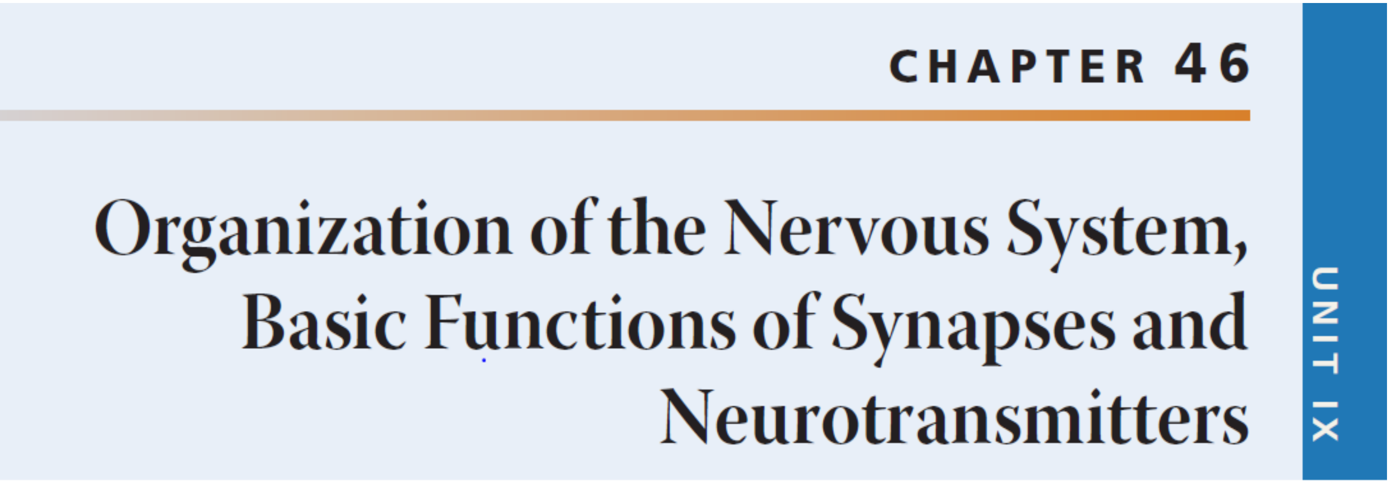
# Brain Computer Interface (BCI)


CHAPTER 46

Organization of the Nervous System, Basic Functions of Synapses and Neurotransmitters

UNIT IX

BGE 2007 -Human Physiology
Lecture-22
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# Chapter Overview:

- Overview of the brain
- Structure and function of central nervous system (CNS) and peripheral nervous system (PNS)
- Structure of the neurons
- Different types of neurotransmitters
- The neural circuits.

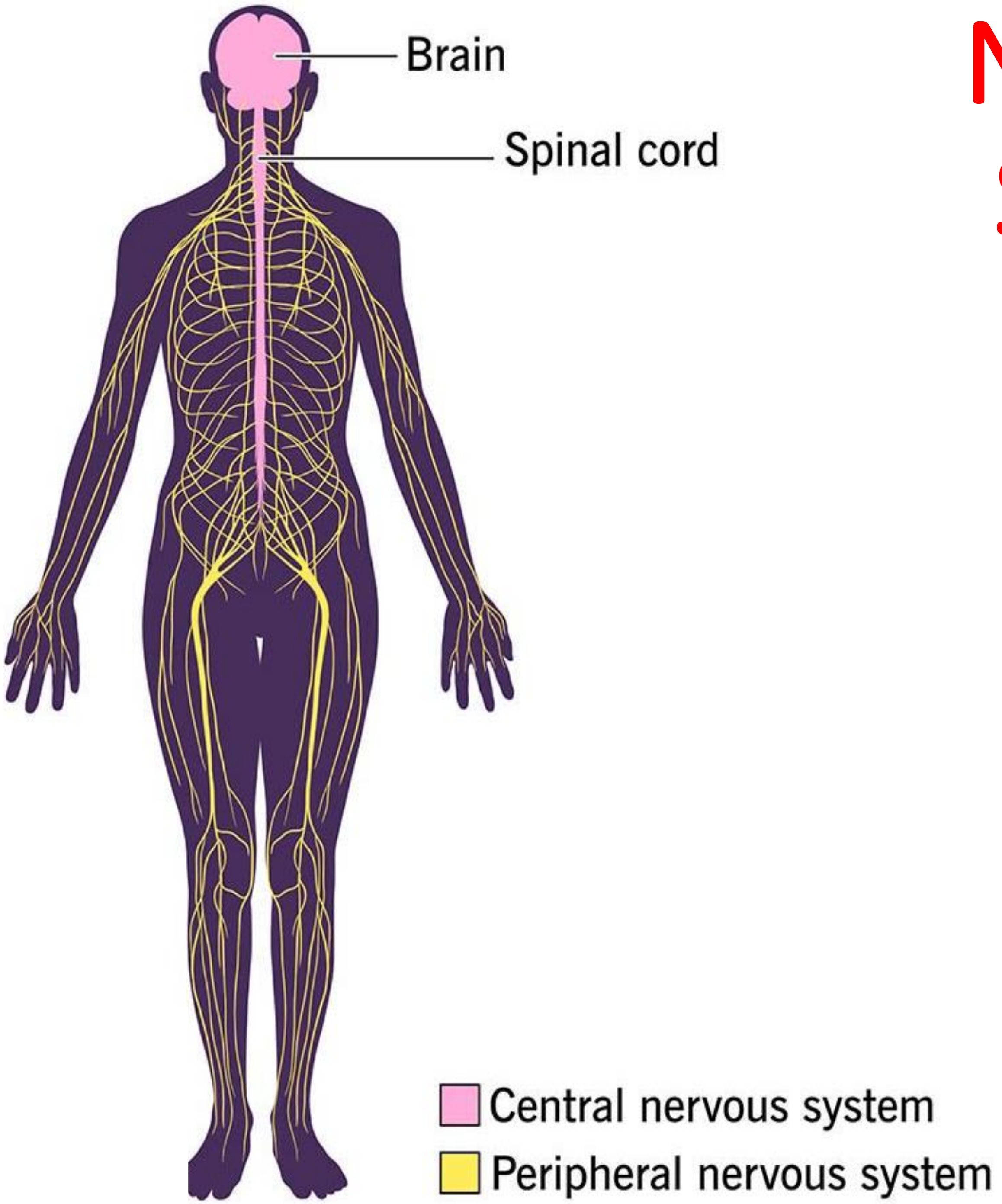
# What is Neuroscience?

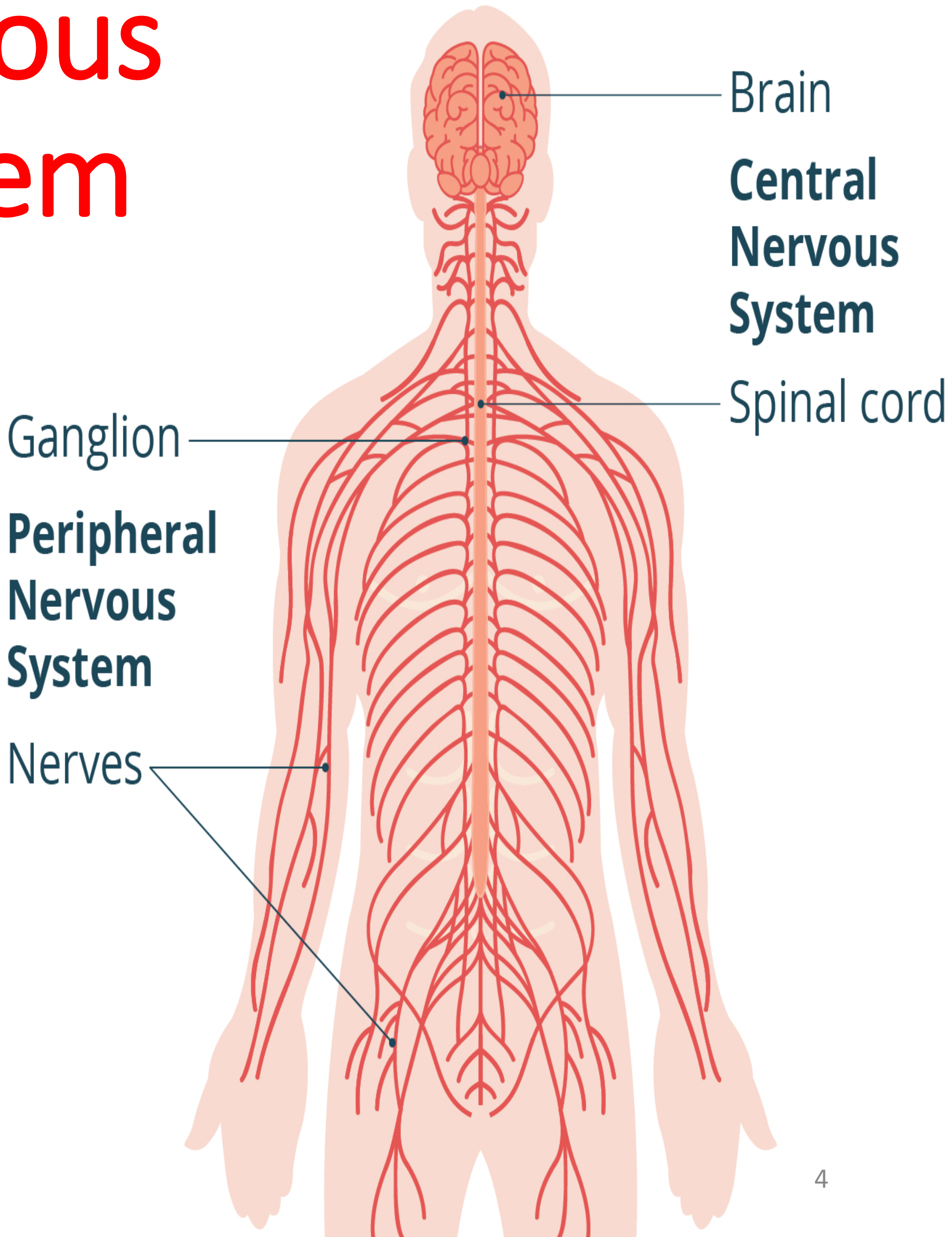
Neuroscience is the study of the **nervous system**, in particular the Central Nervous System (CNS).

**The CNS** consists of **the brain** and **spinal cord** – i.e. the organs that allow us to *make a decision (the brain)* and to *relay that decision to the body (spinal cord).*

**The Peripheral Nervous System (PNS)** carries instructions for actions from the spinal cord to effectors organs (organs that actually do the work e.g. muscles).

# Nervous System

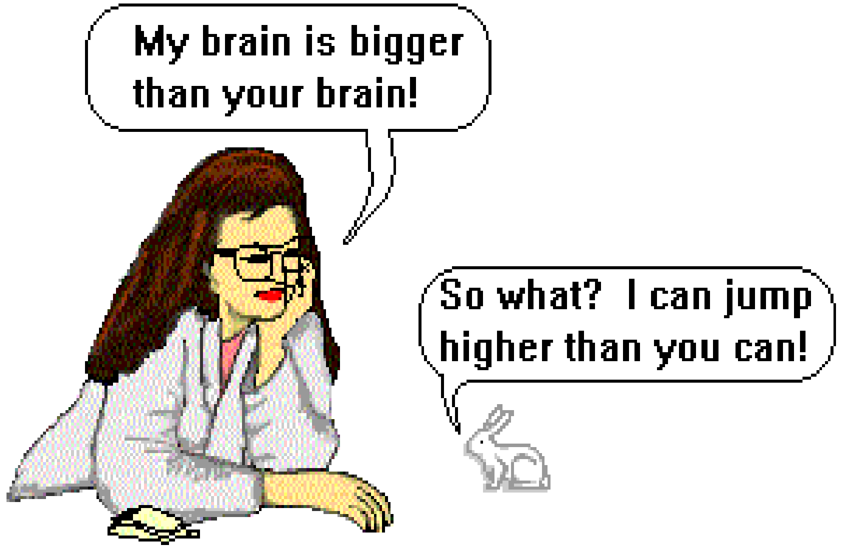


# The Central Nervous System

- The **brain** and the **spinal cord** are the central nervous system, and they represent the main organs of the nervous system.
- The spinal cord is a single structure, whereas the adult brain is described in terms of four major regions: the **cerebrum**, the **diencephalon**, the **brain stem**, and the **cerebellum**.

# Does bigger brain size include higher intelligence?

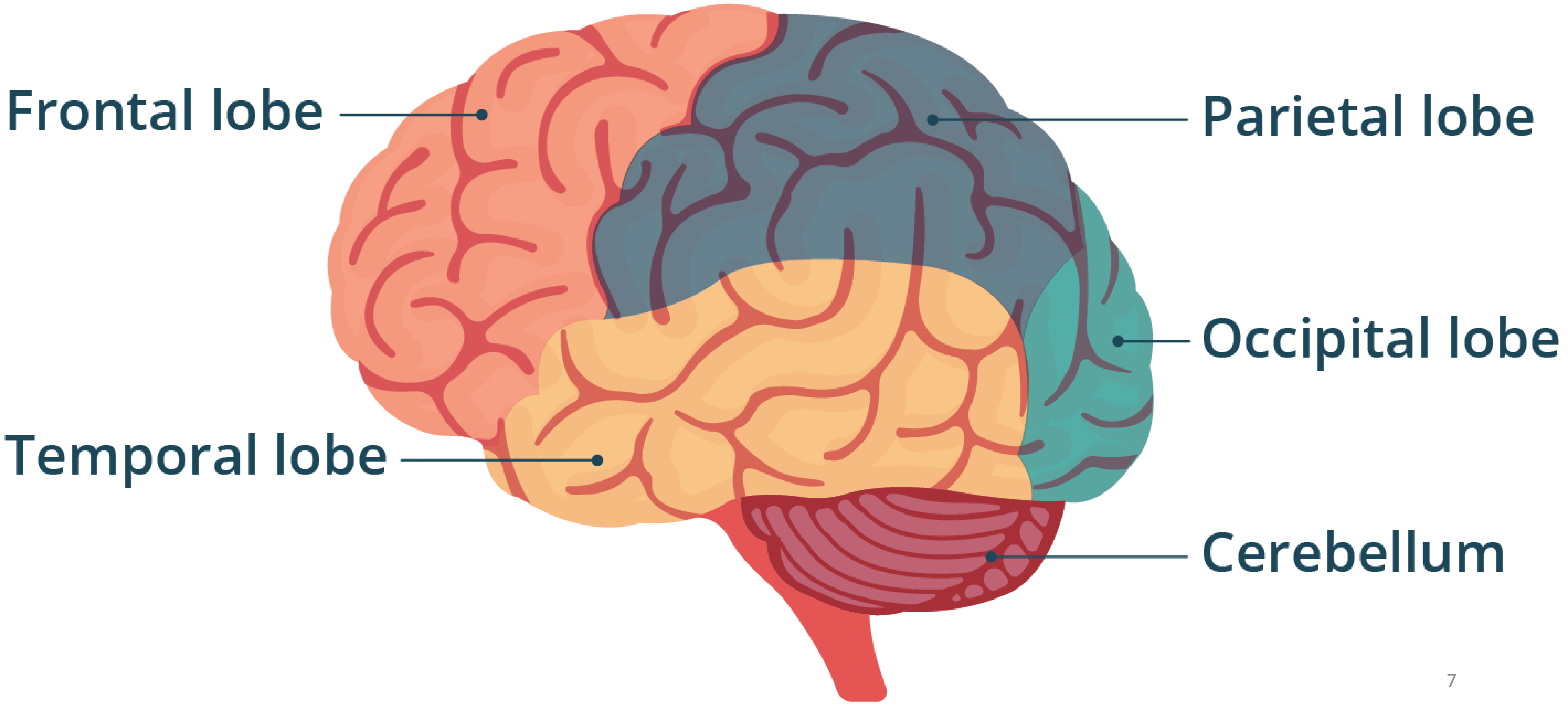



Larger animals  have larger brains, but understanding the relationship is important for two main reasons:

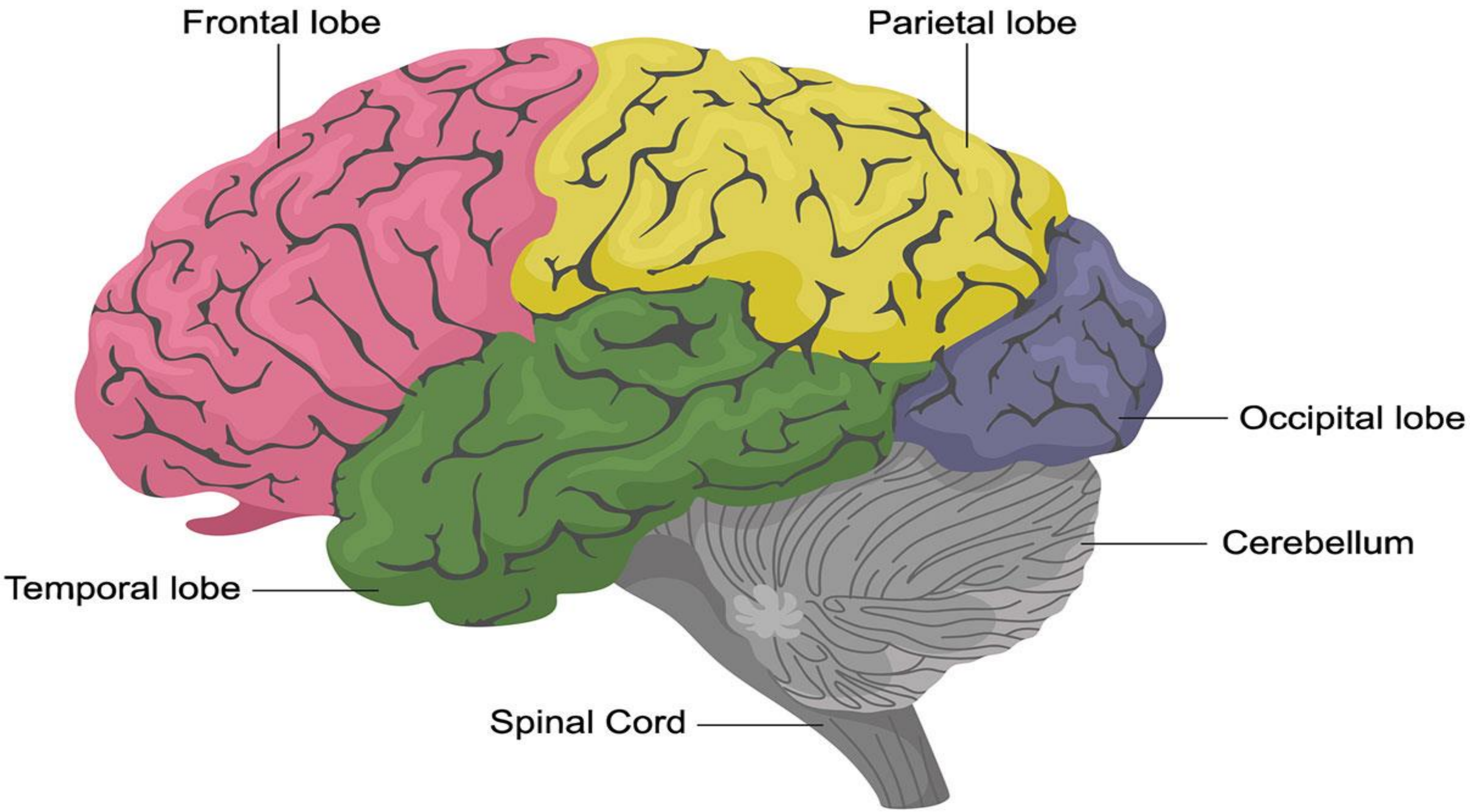
For example, cows have bigger brains than most monkeys, but that most likely has more to do with the having larger bodies than large cow intelligence.

Several factors  determine the higher intelligence. This is why we need to know the structure and function of the brain

# Major brain Areas:



# Human Brain Anatomy



# FRONTAL LOBE

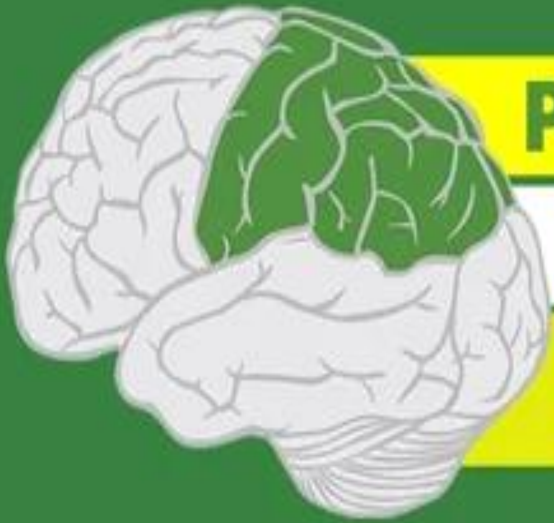
The frontal lobe is the home of much of what makes us human. It plays a role in everything from movement to intelligence, helps us anticipate the consequences of our actions, and aids in the planning of future actions.

## IT'S INVOLVED IN A WIDE RANGE OF "HIGHER" COGNITIVE FUNCTIONS

- ASSESS FUTURE CONSEQUENCES OF ACTIONS
- UNDERSTAND LANGUAGE
- COORDINATE VOLUNTARY MOVEMENTS

## MORE IMPORTANT FUNCTIONS OF THE FRONTAL LOBE

- EMOTIONAL EXPRESSION
- MANAGING REWARD
- ATTENTION REGULATION



# PARIETAL LOBES

The parietal lobe rests near the top and center of the cerebral cortex, just behind the frontal lobe and above the occipital and temporal lobes.

## IT IS VITAL FOR SENSORY PERCEPTION AND INTEGRATION

- SIGHT
- TASTE
- HEARING
- SMELL
- TOUCH

## SOME IMPORTANT FUNCTIONS OF THE PARIETAL LOBE

- COGNITION
- SPEECH
- WRITING

# OCCIPITAL LOBE

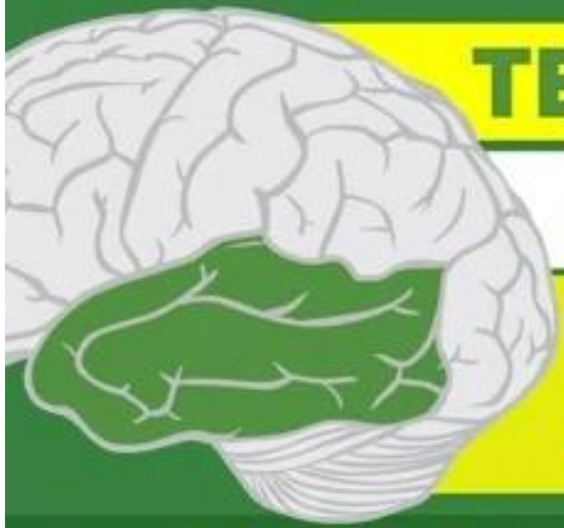
The occipital lobe is the seat of most of the bra
visual cortex, allowing you not only to see and
process stimuli from the external world, but als
assign meaning to and remember visual percep

## IT IS IS ALSO DEDICATED TO VISION, SOME HIGH COMPLEX FUNCTIONS INCLUDE:

- ASSESS DISTANCE, SIZE, AND DEPTH
- MAPPING THE VISUAL WORLD
- DETERMINING COLOR PROPERTIES

## MORE IMPORTANT FUNCTIONS OF THE OCCIPITAL LOBE

- RECOGNIZE OBJECT MOVEMENT
- READING
- IDENTIFY VISUAL STIMULI



# TEMPORAL LOBE

The temporal lobe is so named because of its
proximity to the temples. It is positioned toward t
base of the center of the cortex, just behind the
temples. Like all other brain regions, it is not a
standalone organ.

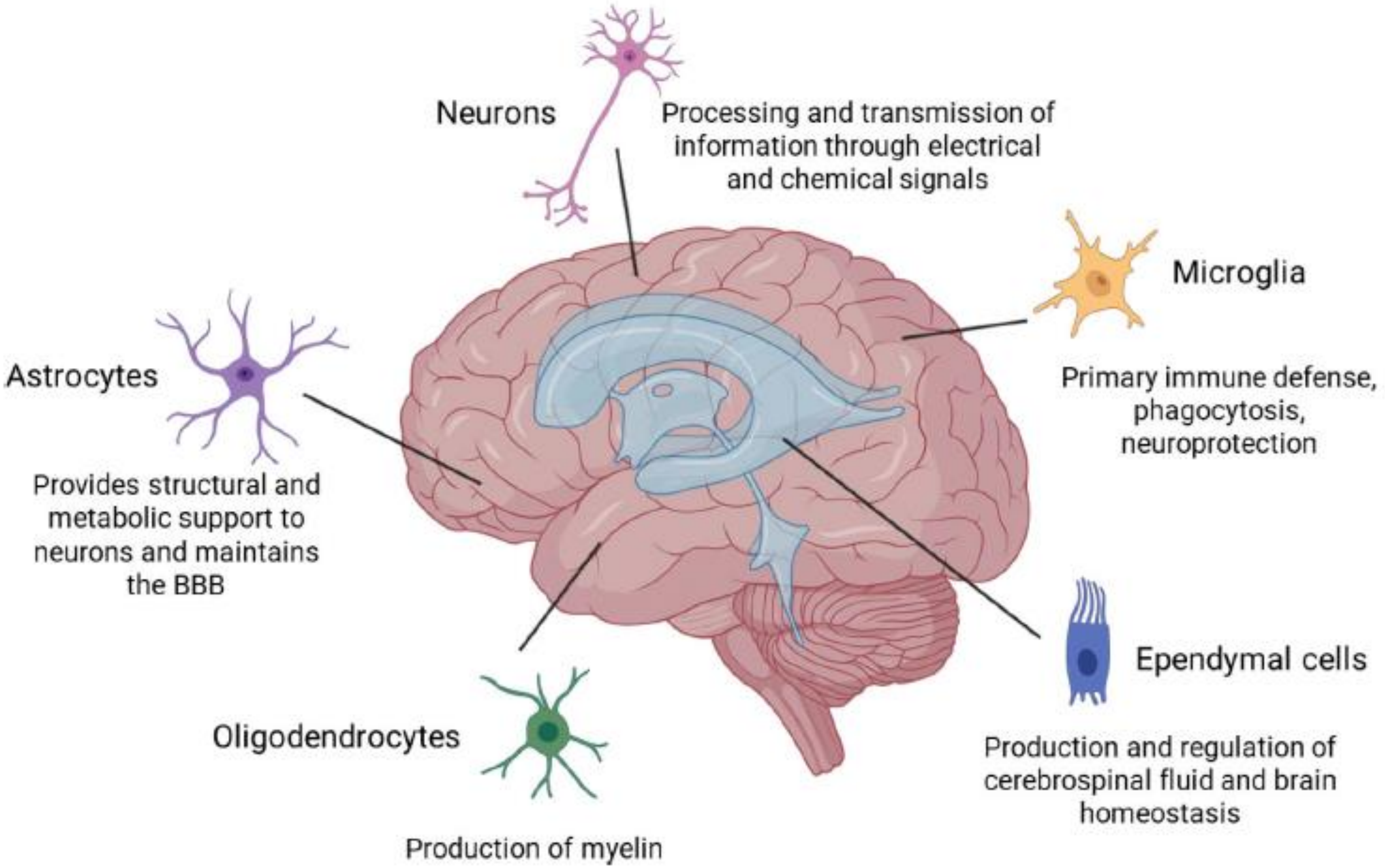
## IT PLAYS A KEY ROLE IN AUDITORY PROCESSING

- PRODUCTION OF SPEECH
- SPEECH COMPREHENSION
- RECOGNIZE LANGUAGE

## MORE IMPORTANT FUNCTIONS OF THE TEMPORAL LOBE

- OBJECT RECOGNITION
- FORMATION OF VISUAL MEMORIES
- MEMORIZING SPEECH

# Major brain cell types:




Major brain cell types: neurons, astrocytes, oligodendrocytes, microglia, and ependymal cells. Neurons transmit signals through chemical and electrical signals. Astrocytes maintain extracellular ion balance, regulate cerebral blood flow, and support the blood–brain barrier (BBB). Oligodendrocytes produce myelin in the central nervous system. Microglia provide immune defense and clean debris. Ependymal cells produce and regulate cerebrospinal fluid and brain homeostasis.

# What is the brain made of?

The bulk of the brain is made up of **structural cells** termed **glial cells** and **astrocytes** and lying in amongst these cells are neurons, specialized cells that conduct electrical impulses along their processes.

It has been estimated that the average human brain contains about **100 billion neurons** and,

on average, each neuron is connected to 1000 other neurons.

This results in the generation of ***vast and complex neural networks*** that are the mainstay of the brain's processing capabilities.

# Neuron

Neuron or nerve cell is defined as the structural and functional unit of nervous system

Neurons are similar to any other cell, having nucleus and all the organs in cytoplasm; except two ways:

1. Neurons has branches called **axons and dendrites**
2. Neuron doesn't have ***centrosome; hence it cannot undergo division***.

# Classification of Neuron

**Neurons**

**Based on the number of poles**

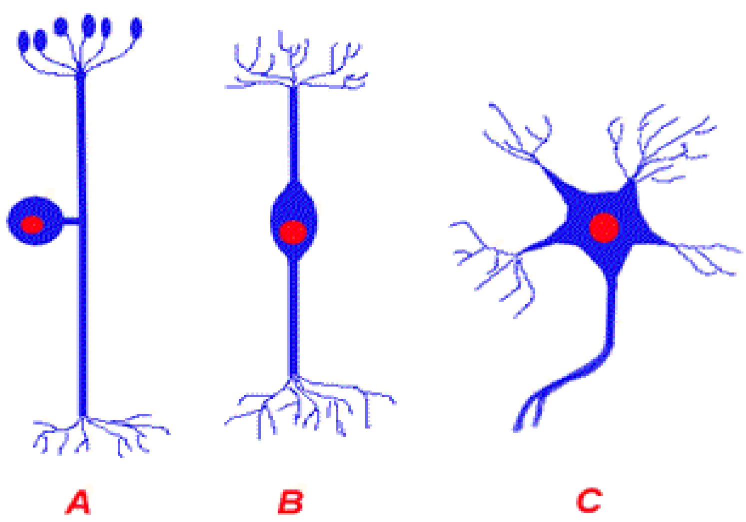
1. Unipolar Neurons
2. Bipolar Neurons
3. Multipolar Neurons

**Based on function**

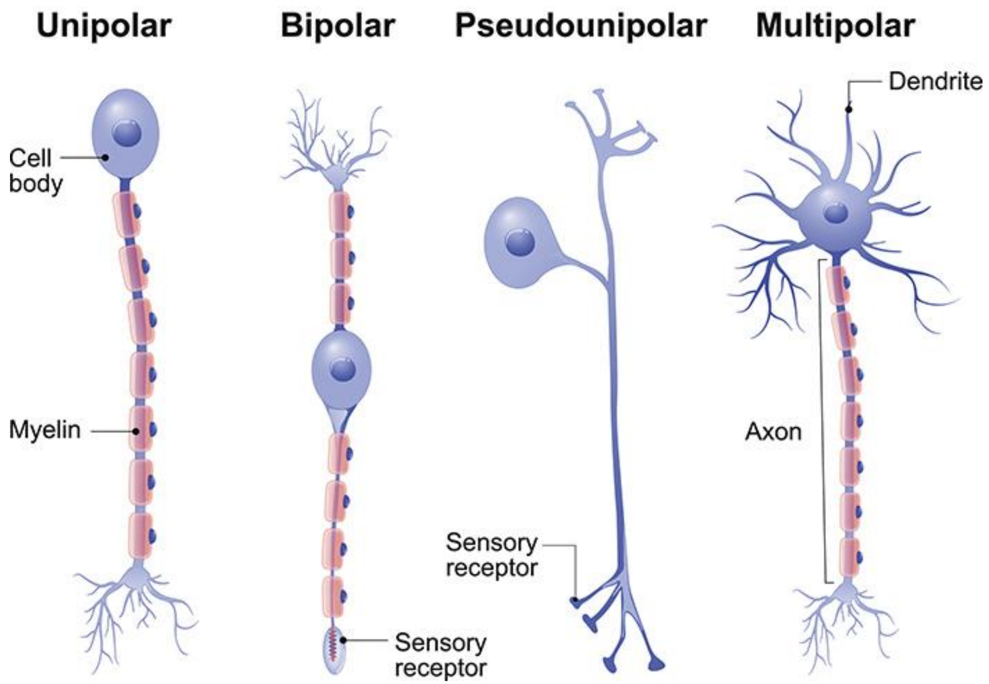
1. Motor or efferent neurons
2. Sensory or afferent neurons

**Length of axon**

1. Golgi type I neurons
2. Golgi type II neurons



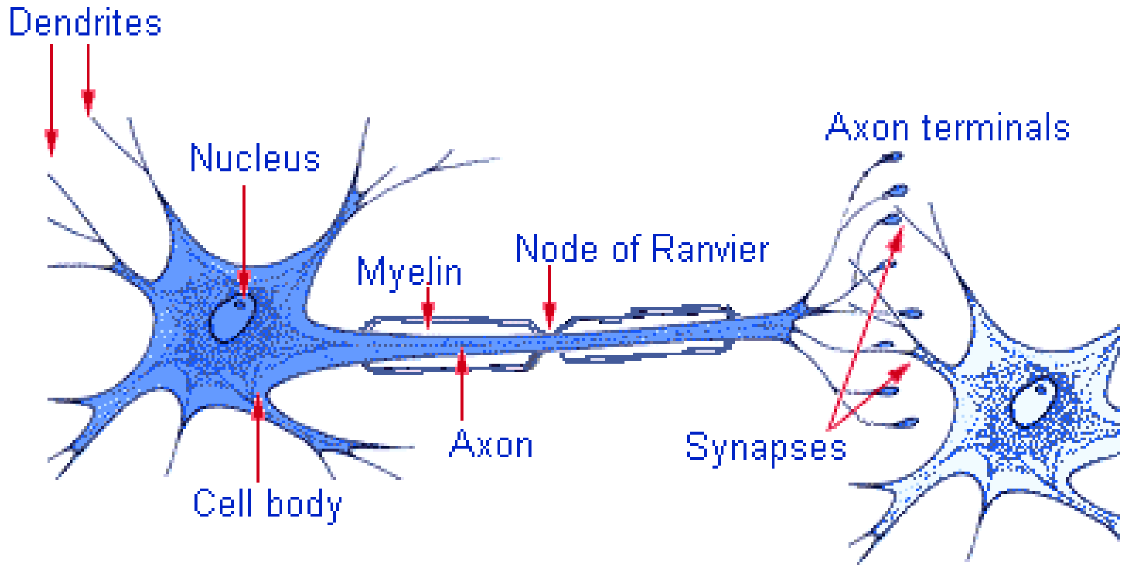

A, Unipolar; B, Bipolar; C, Multipolar

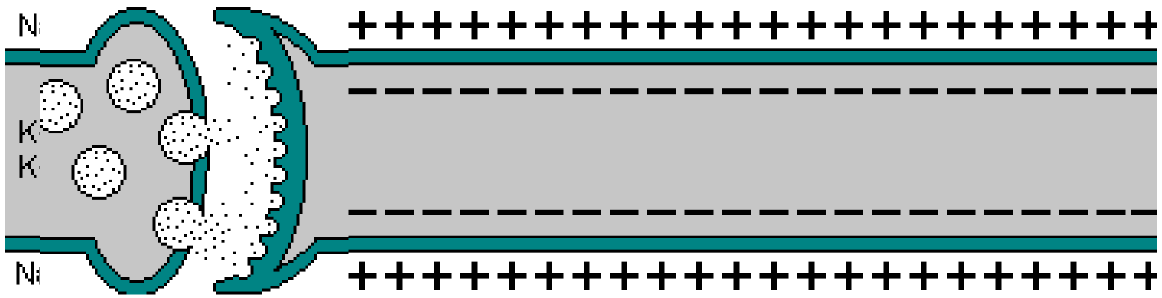



Based on the number of processes that extend from cell body.

- Pseudounipolar: short single process that branches like a T. For example, sensory neurons.
- Bipolar neurons: Have 2 processes. Present in retina of the eye.
- Multipolar: Have several dendrites and 1 axon. Example: Motor neurons.

# Structure of Neuron



- Neurons are the excitable nerve cells of the brain. They are involved in the initiation and conduction of nerve impulses.

- Morphologically, in a typical neuron, three major regions can be defined

(a) The cell body

(b) A single axon

(c) A variable number of dendrites

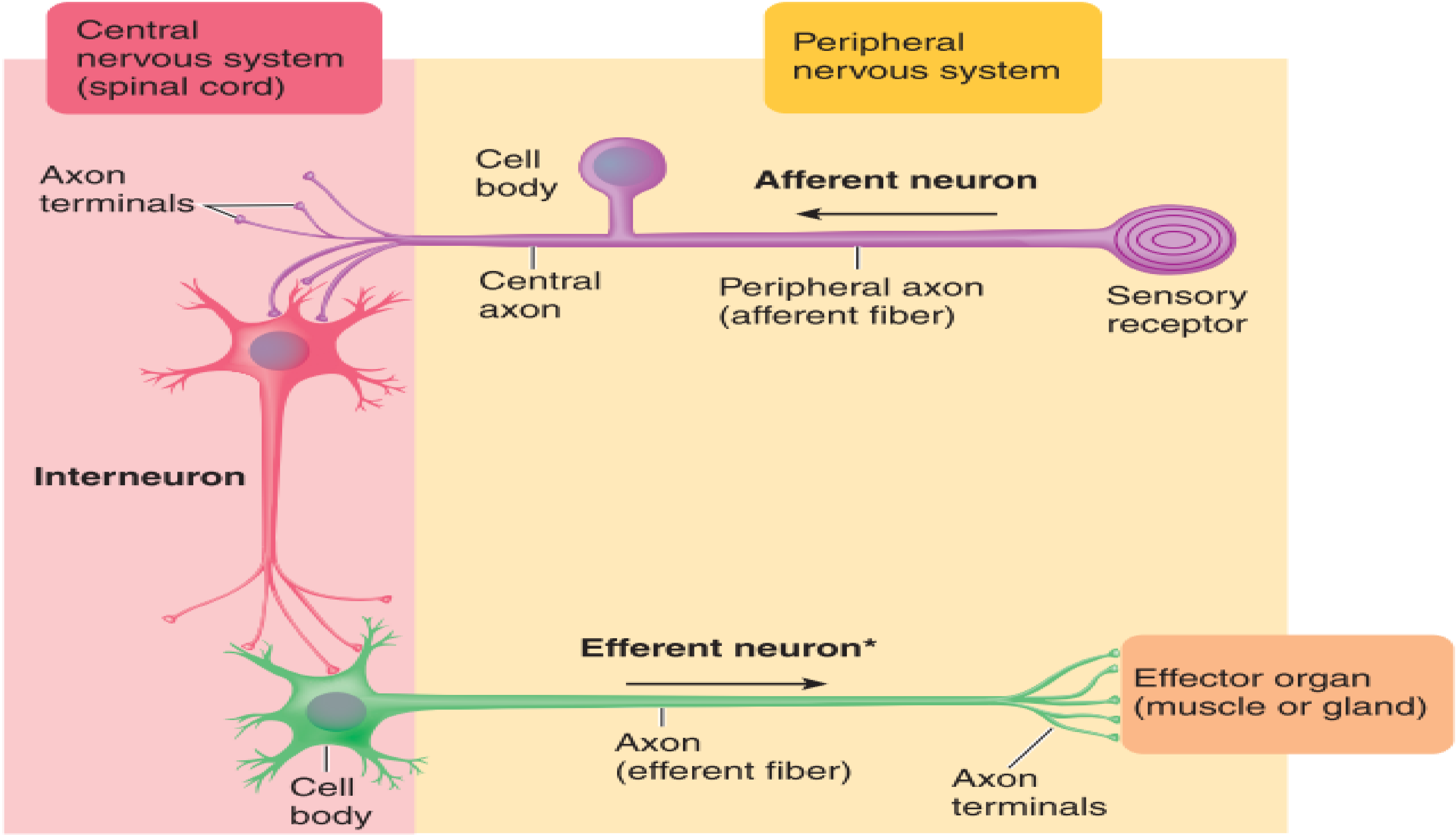
# The nervous system is an electrochemical communication

1. It receives sensory messages from the external environment.
2. It organizes information and integrates it with already stored information.
3. It uses integrated information to send out messages to muscles and glands, producing organized movement and secretions.
4. It provides the basis for conscious experience.

# Neurons/Nerve Cells of the nervous system

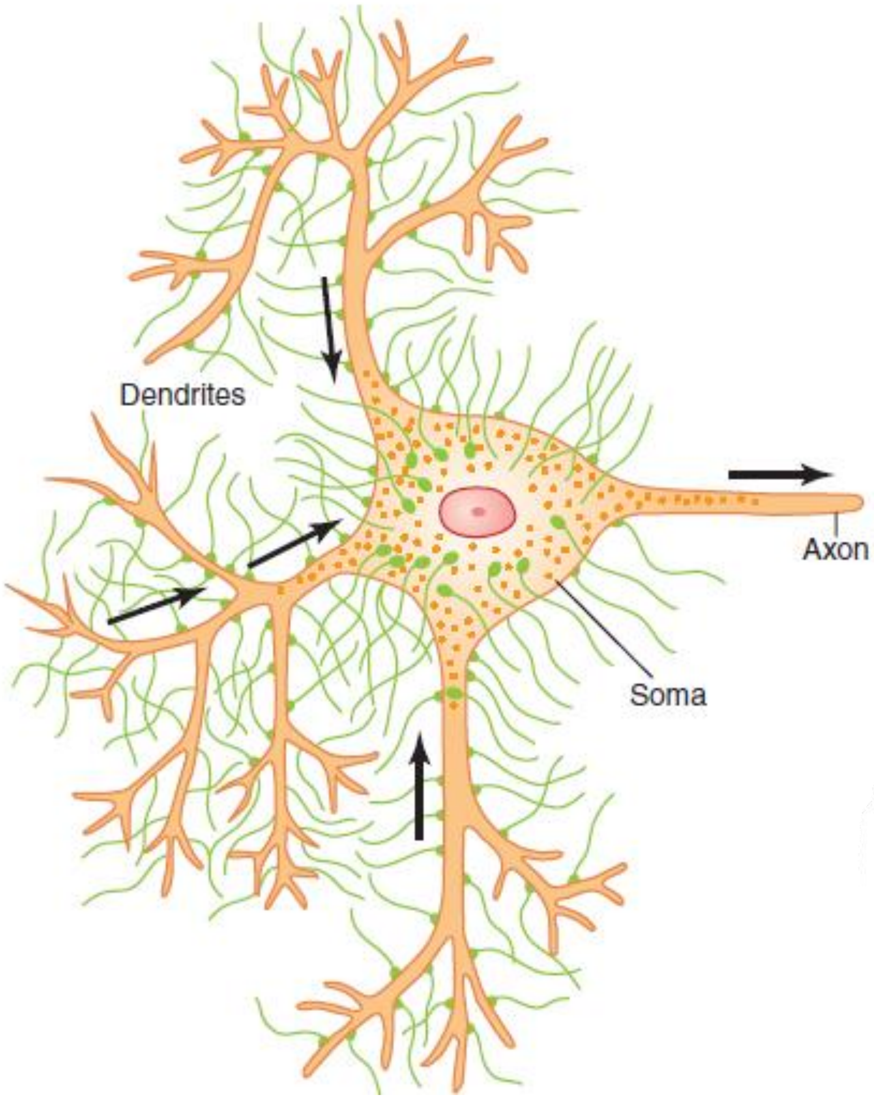
- Nerves are typically **sensory (afferent):** carrying information to the central nervous system from sensory neurons whose cell bodies are located in the periphery of the body.

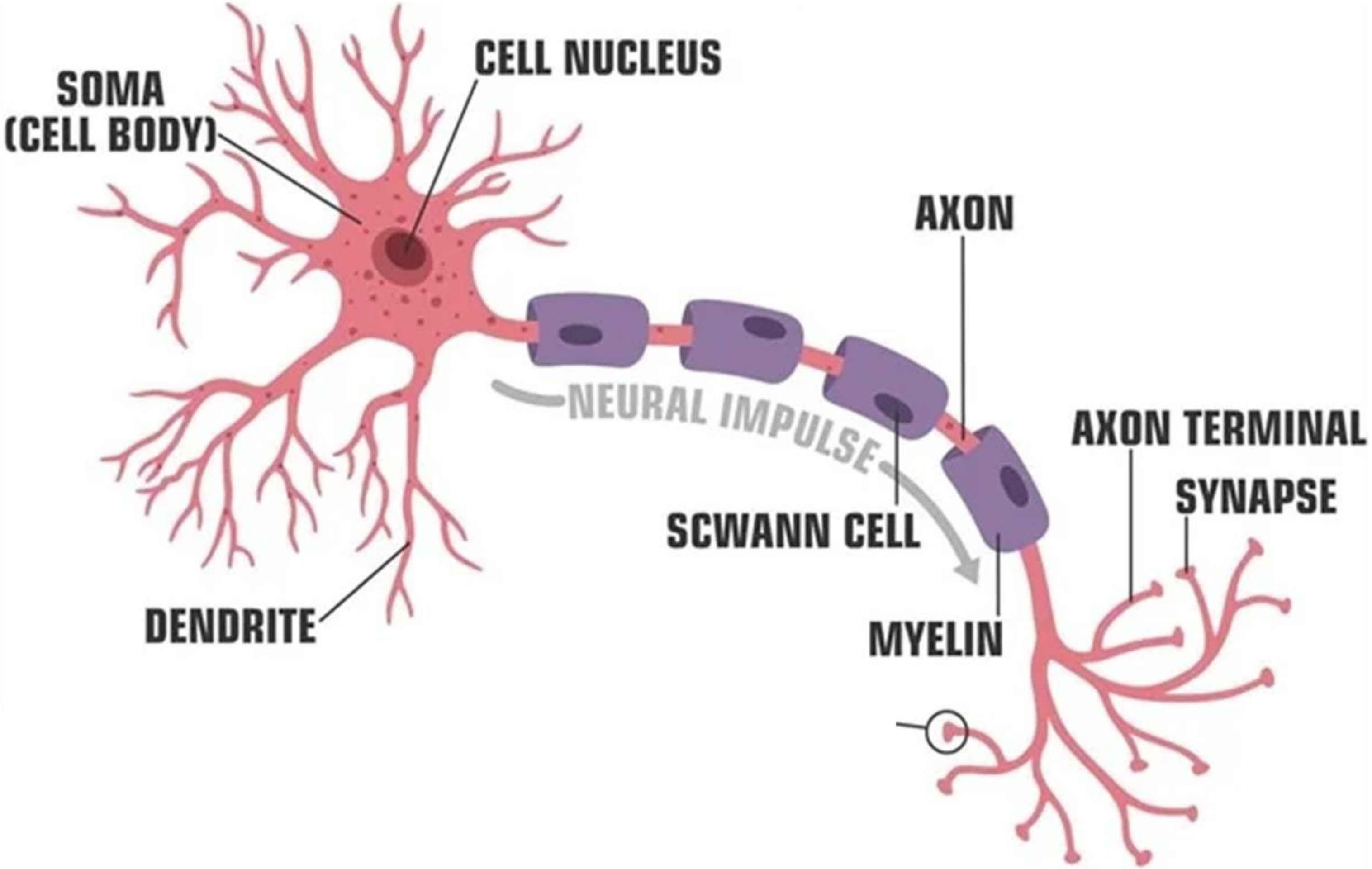
- …or **motor (efferent):** extending out from the central nervous system to the organs and regulating muscular movement or glandular secretion.



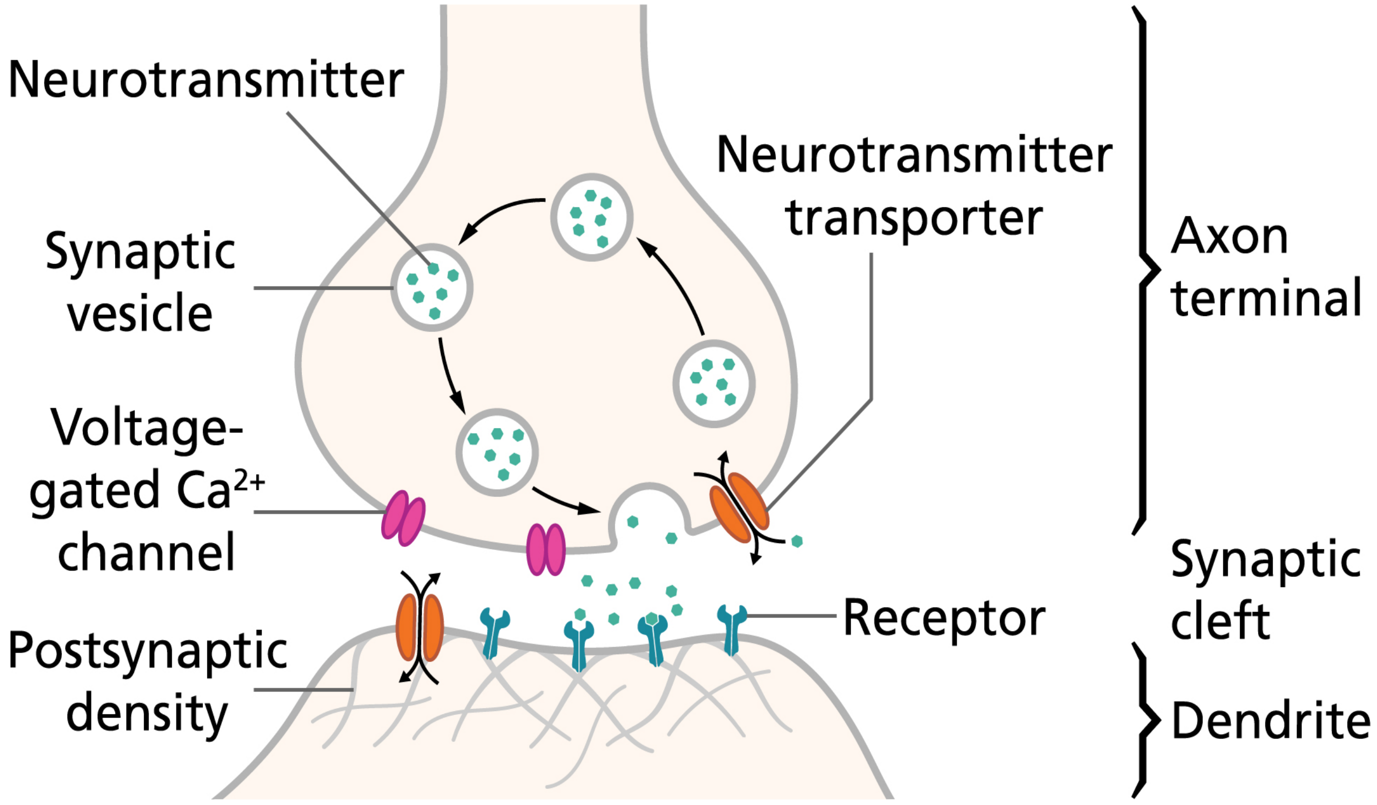
Central nervous system (spinal cord)
Peripheral nervous system
Axon terminals
Cell body
Afferent neuron
Central axon
Peripheral axon (afferent fiber)
Sensory receptor
Interneuron
Efferent neuron*
Effector organ (muscle or gland)
Axon (efferent fiber)
Cell body
Axon terminals

# Neuron

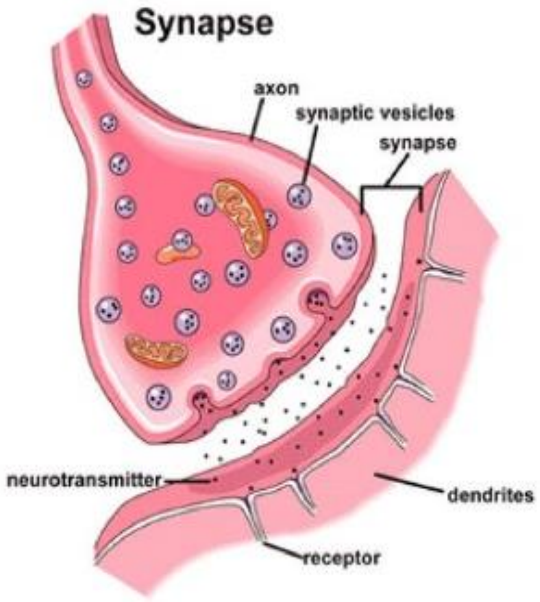

Dendrites
Axon
Soma


SOMA
(CELL BODY)
CELL NUCLEUS
AXON
AXON TERMINAL
SYNAPSE
NEURAL IMPULSE
SCWANN CELL
MYELIN
DENDRITE

# Synapses: how neurons communicate with each other




- Junction between two neurons
- Synaptic Cleft:
  - Gap that separates the two neurons.



# Synapses: Different types

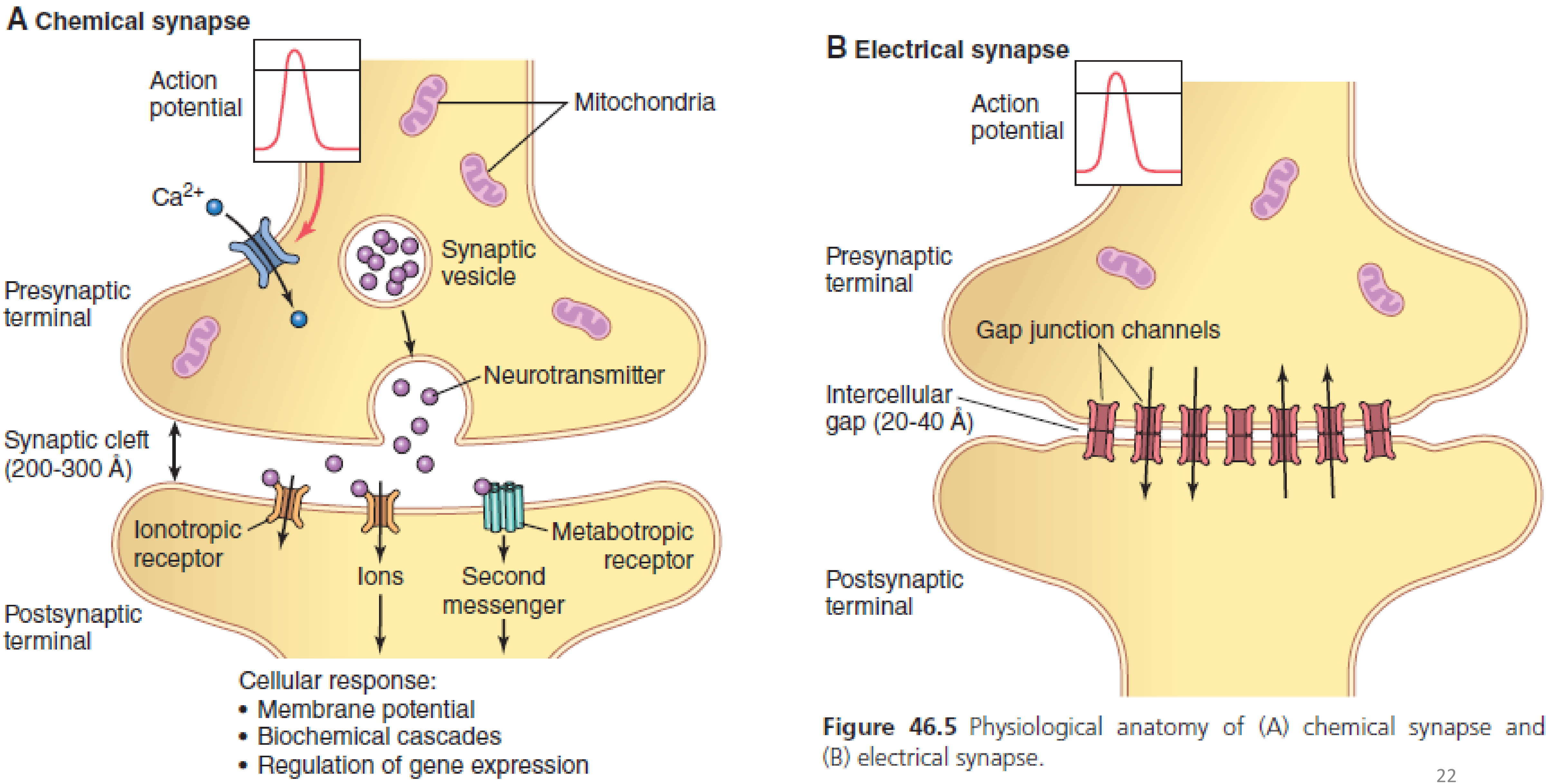
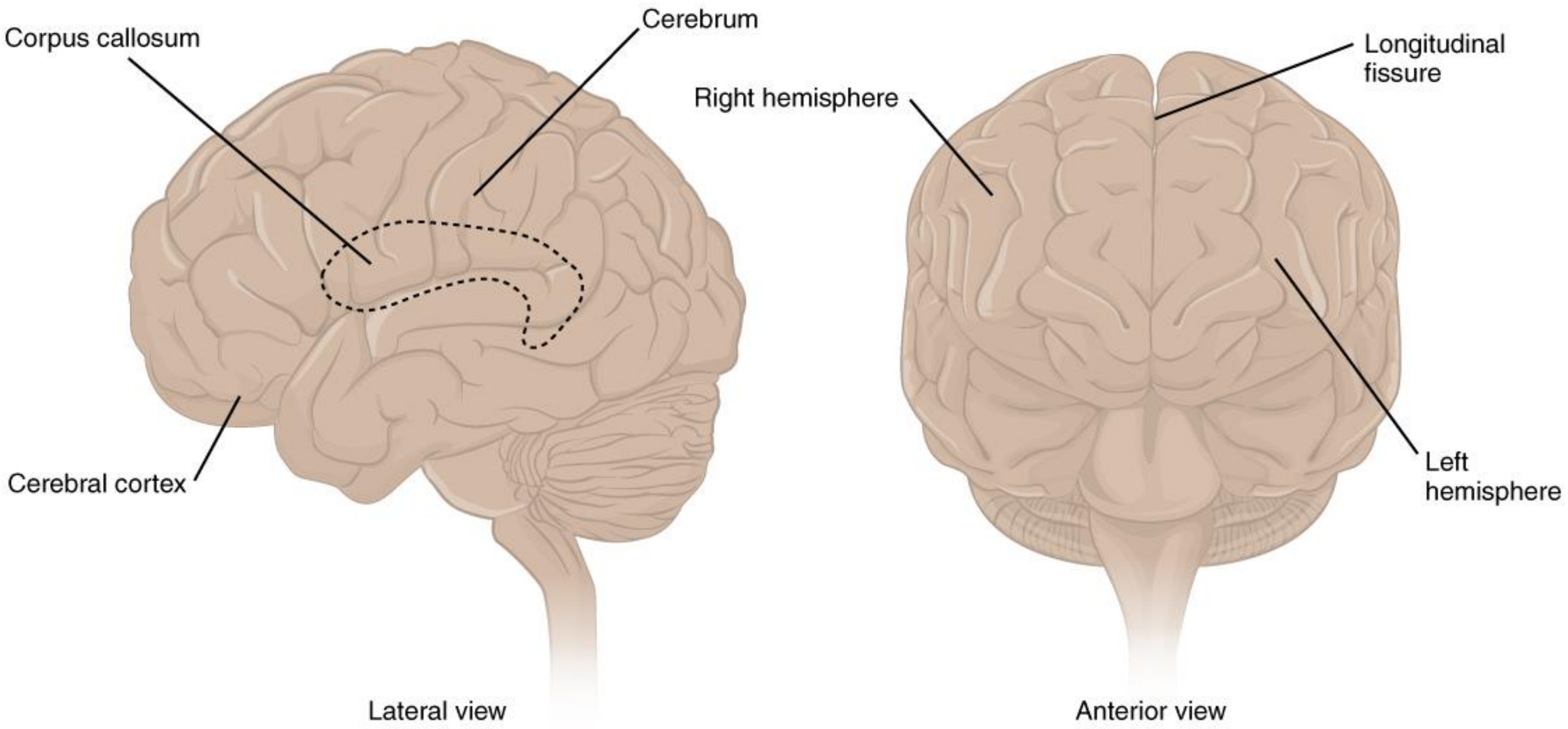


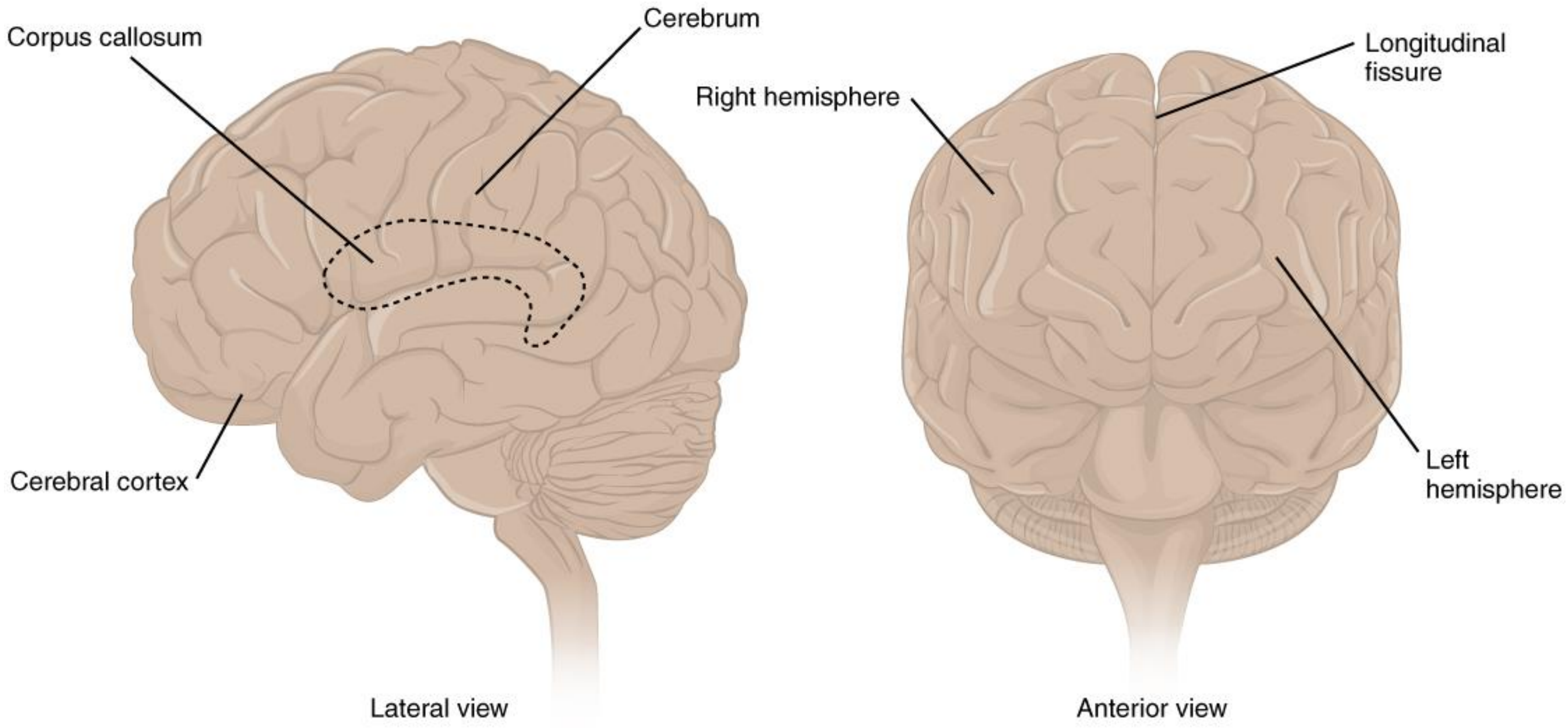

**Figure 46.5** Physiological anatomy of (A) chemical synapse and (B) electrical synapse.

# The Cerebrum



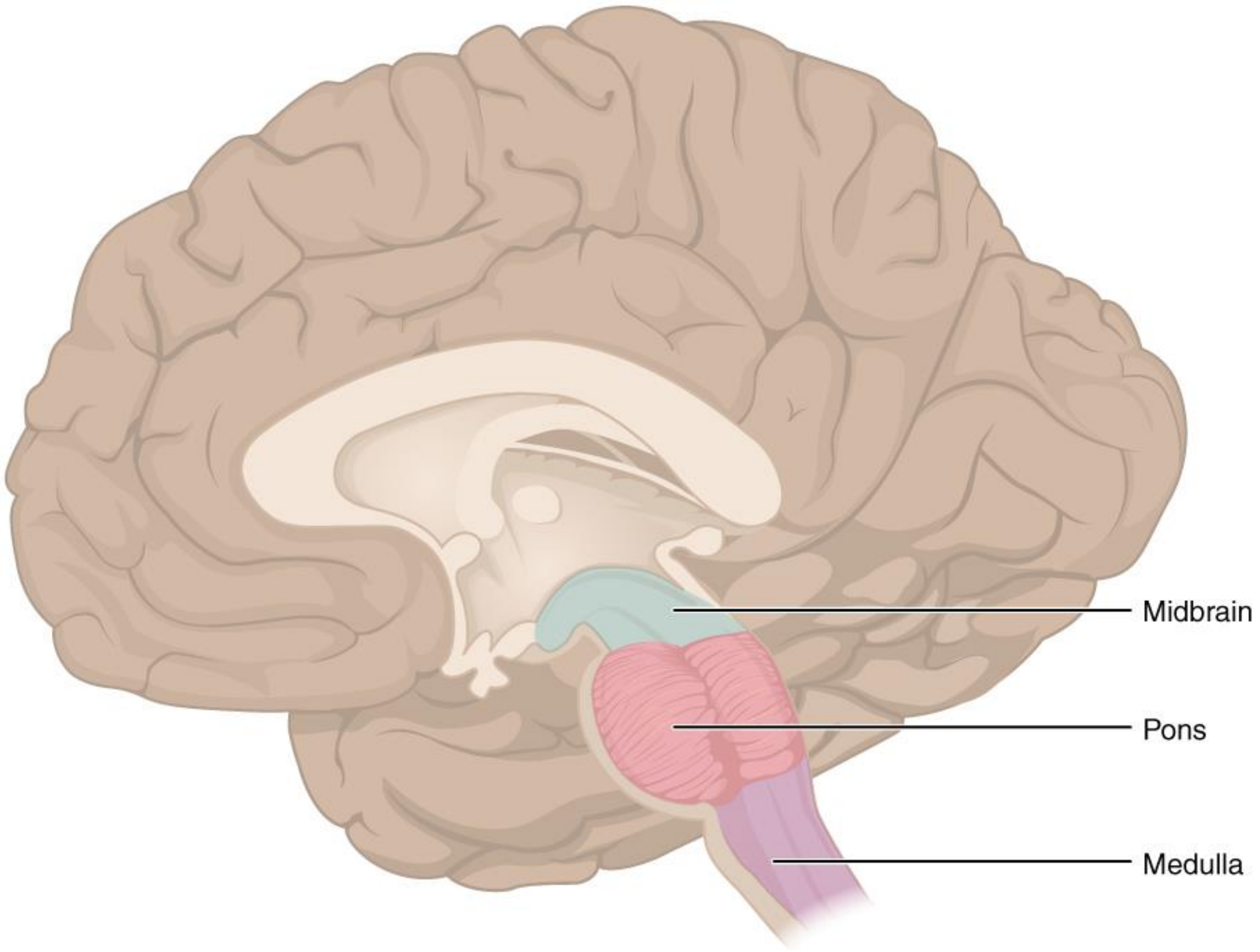
# The Cerebrum

- The **gray mantle** of the human brain that make up most of the mass of the brain
- The wrinkled portion is the cerebral cortex.



# Brain Stem

- The midbrain, the pons and the medulla are collectively referred to as the brain stem.
- The midbrain coordinates sensory representations of the visual, auditory, and somatosensory perceptual spaces.
- The pons is the main connection with the cerebellum. The pons and the medulla regulate several crucial functions, including the cardiovascular and respiratory systems and rates.



# Top Institutes for Learning, Memory, and Cognition

| Country | Strengths | Website |
|---|---|---|
| USA | Learning, memory, Alzheimer's disease, systems neuroscience, computational neuroscience | Picower Institute for Learning and Memory |
| USA | Brain atlases, connectomics, single-cell neuroscience, open science/data resources | Allen Institute for Brain Science |
| USA | Neural circuits, neurodegeneration, genetics, aging research | Salk Institute for Biological Studies |
| USA | Neurotechnology, connectomics, advanced imaging, circuit neuroscience | Janelia Research Campus |
| Germany | Cellular neuroscience, neural circuits, cognition, connectomics | Max Planck Institute for Brain Research |
| Australia | Dementia, memory, neural circuits, neuroplasticity, brain disorders | Queensland Brain Institute |
| Austria | Systems neuroscience, computational neuroscience, neural circuits | Institute of Science and Technology Austria (ISTA) |
| Portugal | Decision making, behavior, neural circuits, cancer and neuroscience research | Champalimaud Foundation |
| Japan | Brain disorders, imaging, neural development, cognitive neuroscience | RIKEN Center for Brain Science |
| China | Developmental neuroscience, genetics, brain disorders, stem cell neuroscience | Institute of Neuroscience, Chinese Academy of Sciences |

# The Cerebrum

- The longitudinal fissure: a separation between the two sides of the cerebrum.
- It separates the cerebrum into two distinct halves, a right and left cerebral hemisphere.
- Deep within the cerebrum, the white matter of the corpus callosum provides the major pathway for communication between the two hemispheres of the cerebral cortex.